Project Design Phase-II

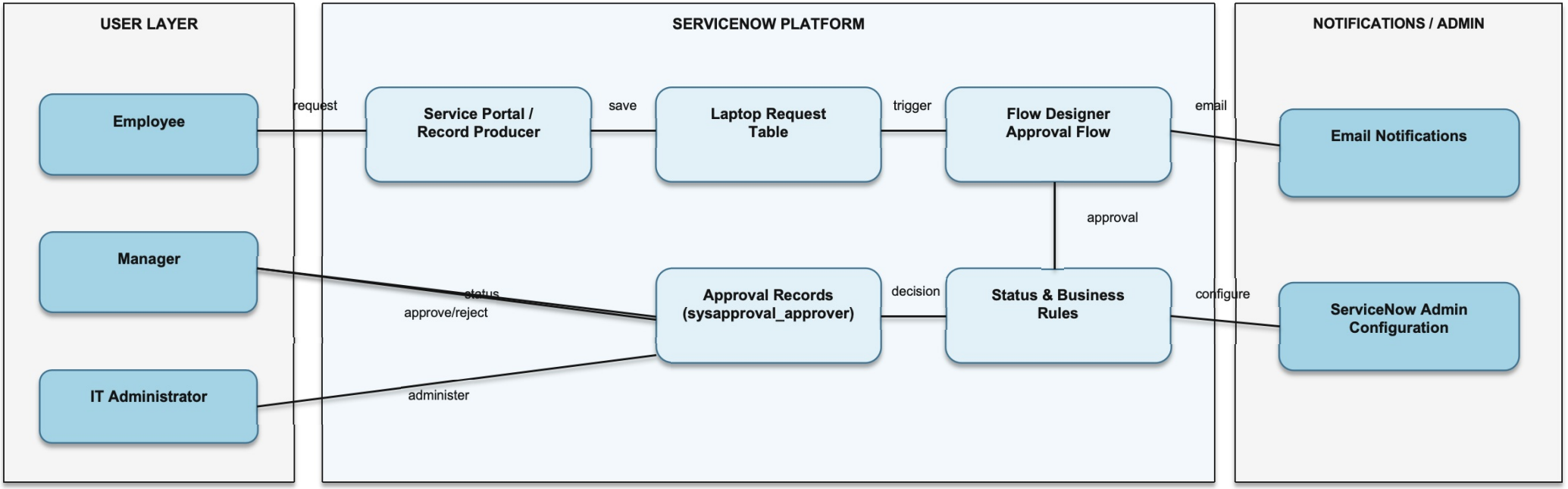
Technology Stack (Architecture & Stack)

Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows the main ServiceNow components and information exchanged during a laptop request.

Architecture Diagram: Employee Laptop Request and Manager Approval



Scope note: Core scope uses ServiceNow platform capabilities. External APIs, cloud databases, machine-learning models and separate file storage are not required unless added to the project.

Table-1: Components & Technologies

Technologies reflect the current ServiceNow-based implementation; optional capabilities are identified.

S.No	Component	Description	Technology
1	User Interface	Employee interface for submitting a laptop request and viewing request information.	ServiceNow Service Portal / Record Producer
2	Application Logic-1	Validates required fields and creates the request record.	Record Producer; platform validation/scripts as configured
3	Application Logic-2	Starts the manager approval workflow after request submission.	Flow Designer
4	Application Logic-3	Checks request conditions and prevents duplicate requests if configured.	ServiceNow Business Rules / GlideRecord
5	Database	Stores request details, requester, status and generated request number.	ServiceNow platform database; custom Laptop Request table
6	Approval Store	Stores the approval task and decision linked to the request.	sysapproval_approver table
7	File Storage	Separate storage is not needed unless request attachments are used.	ServiceNow attachment storage, if applicable
8	External API-1	No external API is required for core request and approval functions.	Not applicable in current scope
9	External API-2	No second external API is required in current scope.	Not applicable in current scope
10	Machine Learning Model	No machine-learning model is required for this rule-based workflow.	Not applicable in current scope
11	Infrastructure (Server / Cloud)	Runs on the organization's ServiceNow instance.	ServiceNow cloud platform; instance configuration

Table-2: Application Characteristics

The characteristics describe the intended design and should match the actual instance configuration.

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses built-in ServiceNow platform components rather than a separately developed open-source framework.	ServiceNow platform; Flow Designer; Record Producer
2	Security Implementations	Restrict request and approval data through roles and access controls; configure ACLs for authorized access.	ServiceNow roles, ACLs, authentication and platform security controls
3	Scalable Architecture	Request and approval logic stays within the ServiceNow platform, which manages its underlying infrastructure.	ServiceNow platform; table-based records; Flow Designer
4	Availability	Availability depends on the organization's ServiceNow instance and its service agreement/configuration.	Hosted ServiceNow instance; platform administration/monitoring
5	Performance	Use efficient queries, appropriate Business Rules and Flows, and avoid duplicate workflow execution.	GlideRecord queries; server-side scripts; Flow Designer

Implementation note: Items marked not applicable are intentionally excluded because the capstone focuses on request submission, manager approval, record tracking and notifications. Only claim features configured and demonstrable in your ServiceNow instance.